

EXP 1: Simulation of FDMA in MATLAB

Aim: To simulate a Frequency Division Multiple Access (FDMA) system using MATLAB and analyze its communication performance in terms of channel bandwidth, data rate, and signal-to-noise ratio (SNR).

Objective 1:

To simulate an FDMA system by allocating **individual frequency channels** to multiple users and determine the **channel bandwidth** assigned to each user.

Procedure :

Open **MATLAB** and create a new script file.

2. Define the total available bandwidth, number of FDMA users, carrier frequencies, signal

power, and noise power.

3. Allocate a unique frequency channel to each user and calculate the **channel bandwidth**.

4. Compute the **data rate** for each user based on the allocated bandwidth and modulation scheme.

5. Calculate the **Signal-to-Noise Ratio (SNR)** using the signal and noise power values.

6. Plot the transmitted signals of all FDMA users using different carrier frequencies.

7. Display the calculated **channel bandwidth, data rate, and SNR** in the MATLAB

Command Window.

8. Analyze the results to verify independent channel allocation and evaluate the overall performance of the FDMA system.

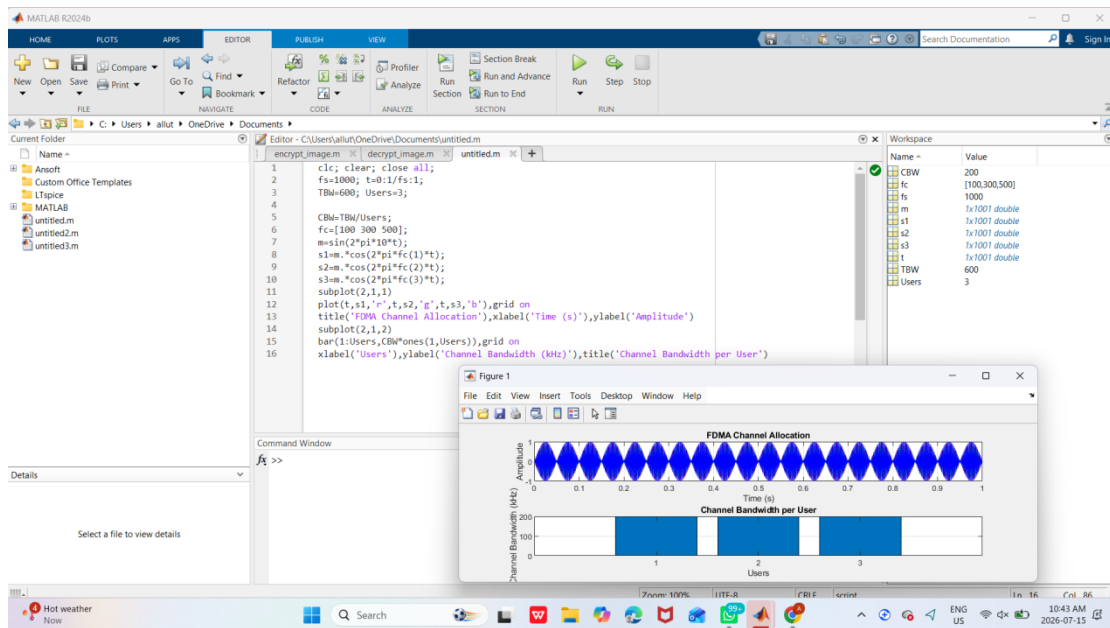
Program (Matlab Code)

```
clc; clear; close all;  
fs=1000; t=0:1/fs:1;  
TBW=600; Users=3;CBW=TBW/Users;  
fc=[100 300 500];  
m=sin(2*pi*10*t);
```

```

s1=m.*cos(2*pi*fc(1)*t);
s2=m.*cos(2*pi*fc(2)*t);
s3=m.*cos(2*pi*fc(3)*t);
subplot(2,1,1)
plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on
title('FDMA Channel Allocation'),xlabel('Time (s)'),ylabel('Amplitude')
subplot(2,1,2)
bar(1:Users,CBW*ones(1,Users)),grid on
xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel Bandwidth per User')

```



Objective 2:

To calculate the **data rate** achieved by each user in the FDMA system based on the allocated channel bandwidth.

Output Parameter: Data Rate (kbps)

Program Code :

```

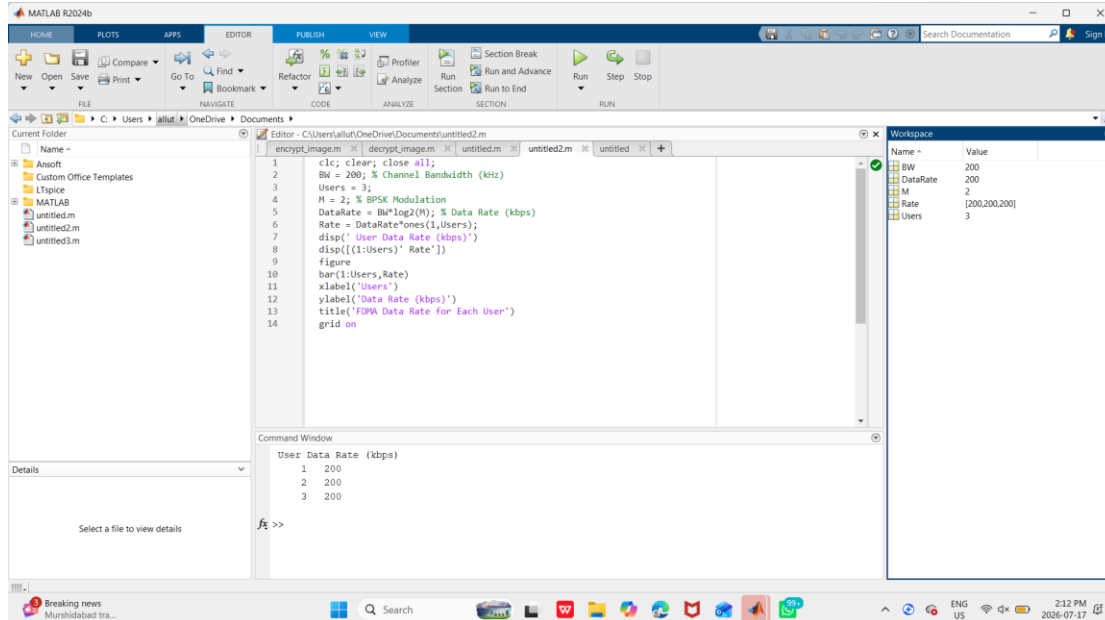
clc; clear; close all;
BW = 200; % Channel Bandwidth (kHz)
Users = 3;
M = 2; % BPSK Modulation
DataRate = BW*log2(M); % Data Rate (kbps)
Rate = DataRate*ones(1,Users);
disp(' User Data Rate (kbps)')
disp([(1:Users)' Rate'])
figure

```

```

bar(1:Users,Rate)
xlabel('Users')
ylabel('Data Rate (kbps)')
title('FDMA Data Rate for Each User')
grid on

```



Objective 3:

To evaluate the **communication quality** of the FDMA system by calculating the **Signal-to-Noise Ratio (SNR)** for different users.

Output Parameter: Signal-to-Noise Ratio (SNR) (dB)

Program Code:

```

clc; clear; close all;
Ps = 1; % Signal Power (W)
Pn = [0.1 0.05 0.01]; % Noise Power (W)
Users = 1:3;
SNR = 10*log10(Ps./Pn); % SNR in dB
disp('User Noise Power(W) SNR(dB)')
disp([Users' Pn' SNR'])
figure
bar(Users,SNR)
xlabel('Users')
ylabel('SNR (dB)')
title('Signal-to-Noise Ratio of FDMA Users')
grid on

```

